

Manual

Extended Planning Information HLS-EPI 8.1

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1 Übersicht Erweiterte Planungsinformationen

Purpose

You use the function package when you require information on

- The capacity utilization of your scheduling
- The assignment of the secondary resources
- The interrelationships in the order network

Integration

The functions are 100% embedded in the graphic planning board. The individual graphic detail applications are displayed on the basis of the scheduled operations.

Features

- Capacity utilization
 - Graphic capacity utilization per workplace, group or cost center
- Utilization profile
 - Graphic capacity utilization profile in the scheduling period
- Histogram
 - Capacity utilization representation in the form of a histogram with colored highlighting of overload.
- Resource view
 - Gantt chart for representation of the secondary resources (tools, etc.)
- Order network
 - Gantt chart for representation of all the operations for an order

2 Utilization Profile

Overview

Menu	Production control → Production preparation → Graphic planning Production control → Production preparation → Info shop floor scheduling
Index tab	Planning details

Usage

You call up the utilization profile of the graphic planning board if you would like to have a chronological overview of how capacities are being utilized in your production department.

Integration

Shown in the utilization profile of the [graphic planning board](#) is a comparison by day, week or month of the *actual assignment* - blue - based on the planned operations and availability (*available resource*) - green - based on the shift calendar. The information is presented in full hours. Generally, hours are rounded down.

Requirement

This requires that you selected a planning profile in the graphic planning board and that you requested data.

Settings

There are the following settings to choose from in the upper area of the utilization profile:

View

This setting controls the chronological grid spacing in the utilization profile. The following settings are available:

Day view: Values are presented for each calendar day beginning at the start of the planning horizon.

Week view: Values are presented for each calendar week. Shown on the X axis is the date respectively for the Monday of a particular week. For the first week, the date for the Monday is also shown, even if this Monday is already in the past.

Month view: Values are presented for each month. Shown on the X axis is the name of the month.



Modifying the performance level for a workplace will not change the utilization, because the available time is calculated using the workplace's shift calendar.

Group

All of the groups for the current planning profile are displayed in the Combo box. When the utilization profile is called up, all groups are highlighted. Because any changes in the selection will affect how the utilization profile is displayed, we recommend that you leave all groups highlighted.

(Quantity)

Number of days/ weeks/ months in the visible area. Pre-assigned when called up: 5.

Editing functions

When the utilization profile is called up, the values for the entire planning profile are displayed, i.e. for all workplaces that have been loaded into the graphic planning board. Clicking on a workplace in the Gantt list of workplaces will display the values for the workplace; clicking on a group will display the values for all workplaces in the group.



For technical reasons, it is not possible to "unhighlight" a workplace or a group. If you would like to display the utilization profile again for the total planning profile, the data must be requested again.

Please also note that if workplaces or groups are highlighted numerous times, only the workplace or group highlighted last will be taken into account.

The size of the utilization profile can be modified by moving the splitter between Gantt and utilization profile.

On the right next to the splitter is a second splitter. This provides the ability to display a tabular workplace list that shows the data workplace, short designation, group and cost center. By highlighting one or several workplaces in this list, you will have the ability to illustrate the utilization profile broken down even further using additional criteria (e.g. by cost center or several groups).

3 Capacity Utilization

Summary

Menu	Production control → Production preparation → Graphic planning Production control → Production preparation → Info shop floor scheduling
Index tab	Planning details

Usage

You call up the capacity utilization in the graphic planning board if you would like to have an overview of how capacities are being utilized in your workplaces or workplace groups over a period of time.

Integration

Shown in the capacity utilization in the [graphic planning board](#) is a comparison of the *actual assignment* - blue - based on the planned operations and availability (*available resource*) - green - based on the shift calendar.

The information is presented in full hours. Generally, hours are rounded down.

Requirement

This requires that you selected a [planning profile](#) in the graphic planning board and that you requested data.

Settings

There are the following settings to choose from in the upper area of the capacity utilization:

View

This setting controls the view and as such how the values are grouped on the X axis. The following settings are available:

- Workplace (Pre-assigned when called up)
- Group
- Cost center

Values are grouped, i.e. how they are presented on the X axis, in alphanumeric order based on setting, workplace, group or cost center.

The information is presented in full hours. Generally, hours are rounded down.



Modifying the performance level for a workplace will not change the utilization, because the available time is calculated using the workplace's shift calendar.

Period

This setting controls the scope of time. The possible alternatives listed below relate to the start date of the planning profile:

- Today (pre-assigned when called up)
- Tomorrow
- This week
- Next week
- Next 7 days
- Next 14 days
- This month
- Next 31 days

(Quantity)

Number of workplaces/ groups/ cost centers in the visible area. Pre-assigned when called up: 5.

Processing functions

When calling up capacity utilization, the values for all of the planning profile's workplaces are displayed. Clicking on a workplace in the Gantt list of workplaces will display the values for the workplace; clicking on a group will display the values for all workplaces in the group.



For technical reasons, it is not possible to "unhighlight" a workplace or a group. If you would like to display the capacity utilization again for the total planning profile, the data must be requested again.

Please also note that if workplaces or groups are highlighted numerous times, only the workplace or group highlighted last will be taken into account.

The size of the detail application can be modified by moving the splitter between Gantt and detail application.

On the right next to the splitter is a second splitter. This provides the ability to display a tabular workplace list that shows the data workplace, short designation, group and cost center. By highlighting one or several workplaces in this list, you will have the ability to illustrate the capacity utilization broken down even further using additional criteria (e.g. by cost center or several groups).

4 Histogram

Overview

Menu	Production control → Preparations for production → Graphic planning Production control → Preparations for production → Info shop floor planning
Tab	Planning details

Purpose

You call the *Histogram* in the graphic *Planning board* if you want to get an overview of the utilization of your workplaces or workplace groups over time.

Integration

The histogram is integrated into the [graphic planning board](#). The histogram displays the *actual assignment* of a workplace over time using the planned operations. To display the histogram of a group, the display is extended and also the operations not yet planned are used.

Requirements

You must select a [planning profile](#) in the graphic planning board and request data.

Editing functions

The histogram is displayed with the Gantt chart of the graphic planning board; it is always displayed below the Gantt.

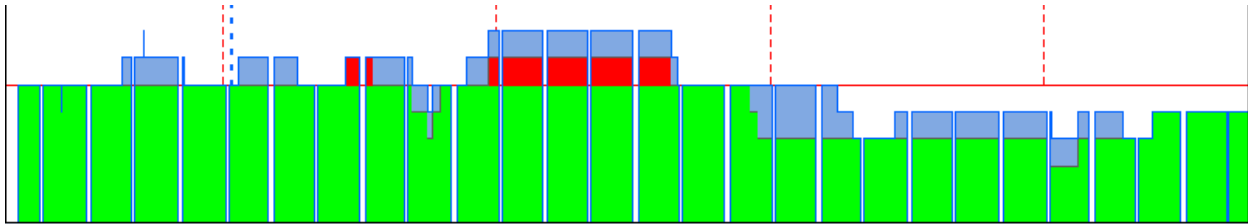
When you call the histogram, the values of the complete planning profile are displayed; this means: the values of all workplaces that are loaded in the graphic planning board. If you click a workplace in the Gantt workplace list, the workplace values are displayed. If you click a group, the values of all workplaces of the group are displayed.

If you replan operations in the graphic planning board (not possible with the *Info shop floor planning*), the histogram display is automatically updated.



For technical reasons, it is not possible to deselect a workplace or group. If you want to show the histogram for the complete planning profile, you must request the data a new time.

Note: If you select several workplaces or groups, only the workplace or group last selected is displayed in the histogram.



Red line

The red line shows the number of available resources (workplaces/machines).

Planned operations

As long as the available capacity is not exceeded, the green area shows the scheduled operations. If the capacity cannot meet the demand, the red line is exceeded and the utilization is displayed in red.

As long as the operations are not yet planned in detail for the workplaces, the required capacity is displayed in blue even if the available capacity cannot meet the demand.

5 Resource View of the Graphic Planning Board

Overview

Menu	Production control → Production preparation → Graphic planning Production control → Production preparation → Info shop floor planning
Tab	Planning details

Usage

You can start the "resource view" in the [graphic planning board](#) to get an overview of the assignment of secondary resources.

Integration

The graphic planning board is used to plan operations onto workplaces. Any allocation of secondary WRM resources (e.g. tools, equipment) is also accounted for in planning.

If an operation requiring a tool is planned, the planner is informed if the tool has already been planned for use elsewhere during the period in question. Yet planners have no overview informing them when the tool is not in use.

The resource view of the graphic planning board also shows assigned resources in a Gantt chart.

Requirement

You can only display such resource types identified as being "relevant for assignment" in the configuration of resource types.

In the "views" tab of the HLS settings you have selected the types of resources which you would like to view in the resource view.

In the graphic planning board you have selected a [planning profile](#) and requested data.



The number of resource types as well as the resources assigned to this resource type affect the time required for requesting data.

Once the resource view has been enabled by clicking the relevant button, it will be displayed and can be placed at the required position within the graphic planning board.

Table

Resources are listed in tabular form on the left-hand side of the resource view. Only resources planned for operations shown in the capacity view are displayed.

When data is requested in the planning board only those resources of operations which have already been planned for workplaces/machines and whose resource(s) is/are available within the Tool and Resource Management module are displayed in the resource view.

Data is displayed in a hierarchical structure: resource type → resource family (name) → resource. Resources that are not assigned to a resource family are displayed in a pseudo resource family.

The below-mentioned data is shown for each resource:

- color of the current resource status
- resource number according to the resource stock
- name of the resource according to the resource stock

The following functions can be started using the context menu of the resource allocation (Gantt):

- Resource configuration (res)
- Maintenance calendar (rmcal)

Gantt

On the right within the resource allocation view, resources are displayed as bars. This area is synchronized with the capacity view. In other words, scrolling into the future in the capacity view results in the resource allocation view being scrolled into the future as well and vice versa; the time scales of the two views are thus always synchronous.

Coloring of bars corresponds to that of the capacity view (according to HLS settings). The bars do not have a color gradient. The MES order number and the article/item number of the operation are displayed as text in the bar.



The resource view does not show any periods outside of shifts; the length of individual resource bars is aligned with the length of the operation; this segment always represents the period between the planned start and the planned end of the operation.

Displaying the resource allocation

If an operation is planned for a workplace and the respective resource has so far not been visible within the resource view the resource will not be displayed. Resource data is not updated automatically. The resource becomes only visible within the resource view after having saved and updated the planning board.

If, as a result of deallocating all of the operations, a resource is no longer allocated at all, it is nonetheless still displayed in the resource view dialog.

Tooltip

The tooltip shows all data selected in the "tooltip" tab of HLS settings. In addition, the workplace and the workplace group are shown for which the operation is planned which the resource has been assigned to.

Context menu

The following functions can be started using the context menu of the resource allocation (Gantt):

-  Order overview (orov)
-  Order information (orin)
-  Deallocating: deallocates the operation which the resource is assigned to.

Displaying blocks

Locked resources are represented by resource bars from the start of the lock to the time when it ends. Locked bars are red, regardless of the color corresponding to the status of the resource labeled as "locked".

Planning functions

The following planning functions are possible in the resource view, provided that the planner has the relevant authorizations for planning.

- Horizontal (chronological) re-planning: By moving a bar in the resource view, the corresponding operation will be re-planned temporally. When "releasing/dropping" the bar, checks are performed according to the settings made in HLS.
- Deallocate: the relevant operation can be deallocated using the context menu.

Both activities result in updating of the other views (Gantt, tabular pool of groups, tabular pool of workplaces).

6 Order Network

Overview

HYDRA menu	Production control → Production preparation → Order network
FEDRA menu	Advanced Process Modeling → Current → Order network
Transaction code	ornet
Function authorization	ornet

Purpose

You use this application if you would like to have an overview of an order network, i.e. a view of linked orders.

Integration

The order network displays order planning results in a Gantt chart.

Requirements

For order networks across orders, you must have defined relationships between operations. The relationships between orders, or more accurately, between operations of related orders, must be transmitted from the ERP system via an interface or

Selection criteria

The application provides the following selection criteria:

Order

The order network for this order can be determined using this field. You can also use wildcards.

When an order number is entered, adjacent orders are selected as well. If the adjacent succeeding orders have predecessors, only their last operation is displayed. If the adjacent preceding orders have successors, only their first operation is displayed.

Project number

If a selection is made by project number, all orders are determined that are assigned to the project number entered. This requires that the project number is defined in the order. You can also use wildcards.

Sales order

If a selection is made by customer sales order, all orders are determined that are assigned to the sales order entered. This requires that the sales order number is defined in the order. You can also use wildcards.

Customer name

If you enter a customer name, all orders are determined that are assigned to the customer name you entered. This requires that the customer name is defined in the order. You can also use wildcards.



Note: Only operations are selected and shown that are at least connected to one other operation and have a planned start or planned end. The application does not display operations of orders including only one stage (i.e. orders with one operation only) that are not connected to an operation of another order. Also operations with an identical planned start and end are not displayed.

Generally, only such orders are integrated that have planning identifiers configured for the order type with either F (detailed planning) or T (scheduling).

Toolbar



Zoom out

Reduces the display.



Zoom in

Enlarges the display.



(Grouped by) sales order

Groups the orders and operations by
sales order → order → operation



(Grouped by) project number

Groups the orders and operations by
project number → order → operation



(Grouped by) order

Groups the orders and operations by
Order → Operation

Detail application

Orders are listed on the *left* in table form. Data can be shown or hidden with groupings . Orders are displayed in groups:

- Project number or customer sales order number (depending on how they are grouped)
 - Order
 - Operation

Displayed at the lowest level are operations with the following information:

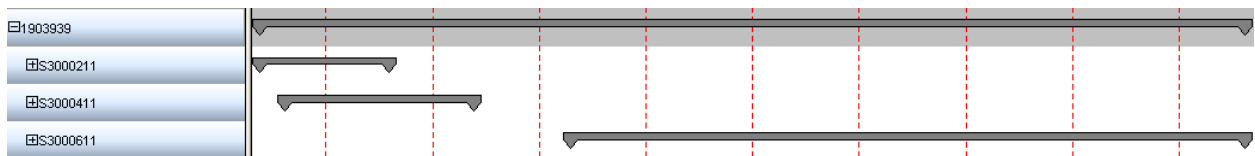
- Operation number
- Workplace at which the operation is planned (if planned)
- Group in which the operation is planned

Operations are shown in the form of a bar on the *right*. For every entry (each line) on the left, there is a corresponding illustration on the right:

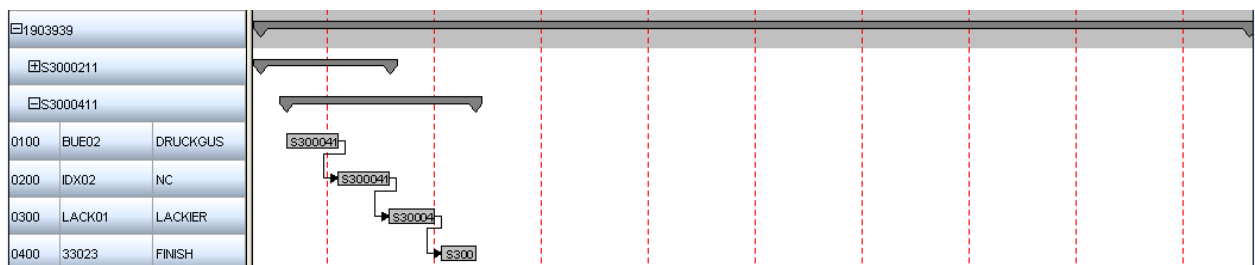
The project order or customer sales order is always displayed in the form of a bar, which is limited at each end with triangles. The length of the bar depends on when each of the orders listed under the project order or sales order, or rather their operations, are scheduled.



If you expand the project order or the sales order, you will see each separate order. Likewise for each order, you will see a bar that is also limited with triangles. The length of the bar depends on when each of the individual operations is scheduled.



When you now expand an order, each of the order's operations is displayed in the form of a bar.



What can now be seen relatively clearly here: the length of the order bar depends on the time domain in which the operations are planned.



All orders are expanded when data is requested and when groups are modified.



The format of the date values displayed on the Gantt chart depends on the format specified by the operating system. The client format is not relevant.

No shift-free times are displayed in the application. The length of the bars in each of the operations is based on the Gregorian calendar. It always equals the length of time between the planned start and the planned end. The coloring of the OP bars is based on the order status configuration.

When you move your mouse cursor over an operation bar, a tool tip is shown that includes the following information:

- MES order number
- Article
- Article designation
- OP designation: operation designation
- Target quantity (P): target quantity in primary quantity unit
- Start: planned start according to planning; if the operation is still in the pool of groups, this is the scheduled start
- End: planned end according to planning; if the operation is still in the pool of groups, this is the scheduled end.

The following applications can be called up from the operation's context menu:

- Order information (function authorization: orin)
- Order progress / order overview (function authorization: orov)
- Edit order network (function authorization: ednet)

Request from graphic planning board

If the order network is called from the graphic planning board, the operations displayed in the Gantt chart are updated with the planned dates scheduled in the graphic planning board, because these may differ from those in the database.